Online Event Management System

https://github.com/aiyshwary/Online-Event-Management-System

SQL NoSQL Database Design Normalization

OVERVIEW

A database-driven system for managing events end-to-end, allowing customers to register, browse and book events with discounts, while giving administrators full control over event listings, users, payments and preferences.

IMPACT

Designed and implemented a normalized relational database and NoSQL schema for an Online Event Management System, covering customer registration, booking workflows, discount management, and payment processing.

TECHNICAL WALKTHROUGH

Online Event Management System – Project Explanation

This project is a database-driven Online Event Management System designed to help users register, browse, and book events online, while allowing admins to manage events, users, discounts, and payments efficiently.

Problem it Solves

Managing events manually is time-consuming and error-prone—especially when handling:

i) Multiple event types

ii) User registrations

iii) Discounts and special preferences

iv) Payments and bookings

This system provides a centralized, structured database that automates all these processes.

Key Actors

User

i) Registers using email and password

ii) Browses events

iii) Books events with number of seats

iv) Applies discounts and special preferences

v) Makes payment using multiple modes

vi) Can update or cancel bookings

Admin

i) Logs in with admin credentials

Adds, updates, or deletes

i) Events

ii) Discounts

iii) Special preferences

iv) User details

v) Manages pricing and discount rates

Core Database Modules

User

Stores registered user details:

i) User ID, Email, Username, Password

Users must register before booking any event.

Event Details

Stores complete event information:

i) Event name, type, date & time

ii) Duration and base price

Each event belongs to a specific event category.

Event Discount

Manages discount options:

- **i)** Premium account
- **ii)** Group booking
- **iii)** Coupon-based discounts

Each discount has a rate reduced value.

Special Preference

Optional add-ons for users:

- **i)** VIP seats
- **ii)** Backstage access
- **iii)** Exclusive services

These come with extra charges.

User Booking

Acts as the core linking table:

- **i)** Connects User, Event, Discount, and Special Preference
- **ii)** Stores number of seats booked
- **iii)** Each booking has a unique Booking ID

This table enforces referential integrity using foreign keys.

Amount

Handles payment details:

- **i)** Payment ID
- **ii)** Booking ID
- **iii)** Mode of payment

Admin

Stores admin login credentials and permissions.

Database Design Highlights

i) Fully normalized (3NF) to avoid redundancy

ii) Strong foreign key relationships

iii) Uses indexes for faster search

iv) Supports views for business insights

Important SQL Features Used

Queries

i) Simple queries

ii) Aggregate queries

iii) Join queries

Views

Examples

i) Total amount payable by each user (including discounts & extra charges)

ii) Users who registered but never booked

iii) Number of bookings per user

iv) Events sorted by price for easy searching

Indexes

i) Faster event search by type & name

ii) Unique email constraint for users

iii) Faster lookup for bookings, discounts, and prices

Overall Summary

This project implements a relational database–based Online Event Management System that efficiently handles user registrations, event bookings, discounts, special preferences, and payments with strong data integrity and optimized query performance.

KEY DETAILS

1. Designed a fully normalized relational schema (up to 3NF) with entities for Users, Events, Bookings, Payments, and Discounts.

2. Modeled equivalent NoSQL document structures for flexible event-type storage and preference handling.

3. Admin interface supports CRUD operations on events, user management, and discount configuration.

4. Customer portal enables account creation, event browsing, booking with preference selection, and payment tracking.

5. Schema documented with ER diagrams, normalization steps, and a detailed project report.